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PHY 102 — Past Questions & Materials
by Renance

Course: PHY 102
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 100

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Level 100 • Second Semester • PastQuestion

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Paper 2 of 4 — phy 102

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Paper 3 of 4 — PHY 102 PAST QUESTION2

Level 100 • Second Semester • PastQuestion

ws through a 1209 resist leeonst

il of regi 231V (c) 122 V (a) 416V 'stor, determine the r.m.s value of

Rats me Se) ca) 5 2:

In such a Way that the plane yal and area 300cm is pl

receive Maximum fly

' Calculate the change in flow of ch

aced in a uniform m

x. Assuming the pla

arge through it, (a) 40y

agnetic field of Strength

ne of coil is now rotated

€ (b) 30uC (c) 50puC (d)

Configuration js the most commonly used in amplificat; Ircui

t Pilication circuit. (a)

' me oa (b) common base (c) common emitrer (d) none of the above

% : phase angle of an R-L-C CACM is oe, when the inductive reactance is greater than the capacilive teactance. (a) Positive(b) hegative(c) zero (d) 360°

9 Two charges QI and Q

10mm. Caleulate the electric field

Q2 is SuC. (a) $4.5 \times 108V$ (b) 22 x

Calculate-the back Y

ie Which, of the follo

above. —_

12. AfKoft Magnetic material is used to prevent which of the fo'lowin

hysteresis loss(c) eddy current loss(d) amperian loss

13, Calculate the maximun amount of cu

134A (b) 141A (c) 125A (d) 116A 5 aa nc

i inductive circuit i (a) -90° (b) $+90^\circ$ (c) 0° (d) -180°

14. The phase angle in a purely inductive circuit is : (a) , " piel cds inva pone, Hees are

15. A 20uF capacitor has a capacitive reactance of ae ay Loe .

frequency of the a.c source? (a) ae ee Gd ot net ie should move In a magnetic of field 6×1057

16. Determine the velocity at which a given rod « eee (h)220mis (€)-400ms (2) 350mn/6

Por an emf of $6 \times 107^*V$ to be induced in Mt. . sower factor is 0.36? (a) 69° tb) 71° (c) 70° (@) 23°

17. What is the phase angle of an R-L-C eae = ee etic field does not depend on which of the following. (a)

: flowing through a wire rotating in a magn tic flux density (d) none of the above.

18, The current lowing il(b) resistance of the coil(c) magne PSO. LefOndh Colsin@ sndil evan aba.

angular velocity of the coil in a series R-L-C circuit where $R=509\ \Omega$,
19. The average power dissipated is
20 kJ (d) |
ss (a) 5 k (b) 15 k (c) a
of a transformer a V_a heat 1068. ϕ a dielectric
ie ip seuhaae) eeat (ϕ) magnetic flux @) the duct atom? (a) egordinate covalent bond(b) cova
vues, (8) Current 0 of bond exist in a sen'con -
2! Width of the ee ae) all of the above h a 50 resistor and @ 20 mH inductor connect
pend (ϕ) ap anti value of the current per a (c) 4.36 A (€) SSA Fs acres Tens dy
22. Deserrouse te c. ty (a) LILAC ' (a) copper Yaas(b) hys
aaa to a 220 SOHz ac supply (a) | induced em. is known as —
a wie vale a @ weasorm ex dye to : : ics the valence band of a Semiconductor is (a) 3 (b)
'. curve fonsid) Gus Jeakgh a cymber of electeune it
MM At recoup tenpmeaaure the
é Ae . iconductor
Bin 40) 4 om te ahearsrs\$. usar to) intrinsic semicon
, ioe cet general A (a) mower #) exasinnic condussar (0)
_panda ; (cs radio { ly rodervel © aa | a
a fa pare ¥
Bie A meant NT tral particl@
be peerone ecnacaanid «ta which & <ppom
g transformer losses. (a) copper loss(b)
rrent flowing through a Sint! inductor, if it stores 50 joules of energy. (a)
referably use '0 WwW Cc AXL ASsSace of
e i d te allo' th maxnnum p. 3
i ron core 18 P
soft won
: 2 is called a
giechentce! cuirgy mpet t
guevent from 4
itis known as (4)
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aay cage ¥
a} ah ; y ity q " Ec San
ee 2, Tre properties of cose gu) coeur 8) 1 : : ~
plametntcamvese Muy pails
au, LP 5
. . S35 1677 Fis)

= : vs. Soucaciior acts Hse aan (a) conductor (b) metal(c) insulator (

33. A tensioner cas a ic ratio oF 16:] and a cvrrent of +Amps is flowing through its secondan windig
Determine the Primary side current. (a) 0.58A () LSA (c) 4.24 (d) 9.04 |

34_ A particle of charge +19uC is Toving from a point X to Y against a uniform electric field of 4×10^4 NA

Calculate the Kinetic energy of the particles when it rerums ta point Y. Let the distance benveen X and Y\$

300cm. (a) 24×10^{-4} J (b) 133×10^5 J (c) 11×10^5 J (d) 13×10^6 J

35. An ideal conductor wes wound

flows through it. What is th

Lit

ed on an insnlating material such that it has 2 1

© magnitude of the magnetic field it genezates? (a) 0.64T (6) 3.3T (c) 0.54T &

36. Assuming the force of attraction benveen nwo charged particles is 1.7×10^8 N when they are placed 60mm apg

In space. Whar will be the magnitude of force when they are 120mm apari? (a) 11×10^8 N (0) 17×10^5 N
 6.7×10^5 (8) 4.25×10^5 N :

37. Suppose iro P2allel wires camming 2 curment of 3.2Amps each are placed 4mm apart.
Determine th

magnitude of the magnetic force per unit leagth experienced by the two wires, (a)(b)(c)(a)

38. The force in Question 37 is -+ + + + - (8) attractive (b) repulsive(c) neutral (qd) none of the above ;

39. An aluminum wire is carrying 2 current oc 1.6A. Determine the Strength of the magnetic field at a pea

* 007m away from its center. (a) 3.3pT (6) 1.2pT (c) 4.57pT (2) 29nT .

: ' F electrons. While in semiconductor, cored

28. In metals, conduction of curvent is due ta the movement of electrons e

conduction is duc to the movement cf 4

free electrons and holes So Hom oi ARE Wyte 5 © the value ofag

29. The rate of change in the current fiowing through a 20mH inductor is SQOA'see. Determine the value o

induced in the coil. (a) 100V (b) 10V (c) 1000V (4) 1V dels sirens tn sendiconductorsie

30. The diffusion process which involves the random motion of holes and free electrons in tia dEsee ISG

to (a) external pressure(b) thermal agitation (c) potential difference applied (a) all at =e 152

31. The length and area of two similar coils A and B placed near each other are 0.23m and $1.4 \times B$ arel

5 Tespectively, Determine Sie mines fe Sse 8 'sca the coils, if the number of tums of coil A and ik:

40. The capacitance of a Capacitor tends to be... when its plates are Separated in an insulating | compared To whea they are placed in vacuum. (a) largez(b) smaller(c) the same(@) none of the above 3

41. The electric field intensity in a capacitor is $16 \times 10^6 \text{ V/m}$ in vacuum. What will be the magnitude of the electric field intensity when a dielectric material of thickness 5.2 mm is placed between the plates? Assume the dielectric constant to be 5. (a) 320 V/m (b) 440 V/m (c) 100 V/m (d) 225 V/m

42. A wire of length 0.5 m lying along the x-axis carries a steady current of 25 A and it experiences a magnetic force of -67 N at a right angle to its axis. Determine the magnetic field intensity: (a) 5.6 kT (b) 7.3 kT (c) 23 kT (d) 4.8 kT

43. Estimate the area of the each required to build a capacitor of $5.3 \times 10^{-7} \text{ F}$ if the plates are to be separated in vacuum. (a) $2430 \times 10^{-3} \text{ m}^2$ (b) 7 m^2 (c) 0.072 m^2 (d) 0.0072 m^2

44. The magnetic field of the magnet is 12 T at a distance of 4.2 mm . (a) $190 \times 10^{-6} \text{ T}$ (b) $190 \times 10^{-6} \text{ T}$ (c) $190 \times 10^{-6} \text{ T}$ (d) $190 \times 10^{-6} \text{ T}$

45. The initial value of 120 uC in the resistor is 3 m (a) 1.1 m (b) 3 m (c) 3 m (d) 0.5 m

46. If the resistance of a solenoid is $3.9 \text{ }\Omega$ and a current of 9.37 A (a) 33313 J (b) 34 J (c) 0.34 J (d) 37 J flows through it. What is dissipated in form of heat?

47. The amount of c will be π (a) 6.1 secs (b) 6.1 secs (c) 6.1 secs (d) 6.1 secs if the drift velocity is 10^6 m/s . The wire of cross-sectional area 2.0 mm^2 is made of copper (a) $71 \text{ }\mu\text{C}$ (b) $66 \text{ }\mu\text{C}$ (c) $71 \text{ }\mu\text{C}$ (d) $66 \text{ }\mu\text{C}$

48. An alpha particle of 4 MeV enters a magnetic field B whose direction is perpendicular to the plane of the paper. The path of the particle is (a) 0.5 m (b) 0.5 m (c) 0.5 m (d) 0.5 m

49. The resistance of $62.50 \text{ }\Omega$ at 70°C is what? (a) $62.50 \text{ }\Omega$ (b) $62.50 \text{ }\Omega$ (c) $62.50 \text{ }\Omega$ (d) $62.50 \text{ }\Omega$

50. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

51. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

52. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

53. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

54. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

55. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

56. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

57. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

58. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

59. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

60. The material to be used in the wire will have a resistance of $100 \text{ }\Omega$. Which material is best? (a) $100 \text{ }\Omega$ (b) $100 \text{ }\Omega$ (c) $100 \text{ }\Omega$ (d) $100 \text{ }\Omega$

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39

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62.

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67.

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69.

70.

. The period of motion a

ol,

(b) $6.56 \times 10^{-2} \text{ N}$ (g) $9.7 \times 10^7 \text{ N}$ (4)

(77.3 vO 8 5 yy 8 Replaced by an electric tie 345 i

5 pe. - AB) $3.3 \times 10^7 \text{ N}$ (c) BTRUO-EN Soon we ie What will be the force on the electron?

al pull On the electron, cn 03 vlo-8y

RE Lorces js Te largess

: has the least ϕ

: Assuming the eleetmon is

MASNee tote it wil] exp

A0nF CAPAcitor y

(Wd) 3.6

- The magnetic flux

ag UX generated by a 200 tums 1

he : g \ 2 m: x: Af a: . ' Ay nw a) $3 \times 10^3 \text{ i}$

iam spe @inae Gases Sper meter wire when a current of 3.0 A flows through it is. (a)

» -\reetangular shaped eles 1

fue ee Sue es woMagnet 'S to produce a flux density of 4.8 T through its surface. If it generates a
rere ren b. what w im be the dimension of the electromagnet? (Let the length to breadth ratio be
 $4:3$) (a)

-* $1 \times 10^{-6} \text{ m}$ ϕ (b) $0.11 \times 10^{-8} \text{ N}$ (©) $0.5 \times 10^{-3} \text{ m}$ (ad) $0.76 \times 10^{-2} \text{ m}$

(0) $3.5 \times 10^{-29} \text{ N}$ (c) $5.3 \times 10^{-30} \text{ N}$ (d)

Oey Stmagnitude?

NOW moving at an angle a

ue Detience? (8) $97.4 \times 10^7 \text{ N}$ (b) $3.57 \times 10^7 \text{ N}$

aS discharged through a 180k Resistor, estimate

sa 40 a :

133° to the Magnetic field, What will be the value of the

N (c) $61.7 \times 10^{-72} \text{N}$ (d) $55.40 \times 10^{726} \text{y}$

e the time constant, (a) 5.1 (b) 6.5 (c) 7.2

A particle of mass 6.27×10^{-28}

5 kg is moving in a circular orbit at night

LOT and its radius of orbit is 20cm. Calculate: (question 60-63)

about the orbit, (a) $3.51 \times 10^7 \text{ s}$ (b) $4.57 \times 10^7 \text{ s}$ (c) 11.7×10^7

energy possessed by the particle. (a) 17.12103eV (b) 3.07×10^6

angle to a uniform magnetic field of

25° (d) 4.0×10^{255}

eV (c) $9.7 \times 10^7 \text{ eV}$ (4)

The amount of

$3.0 \times 10^7 \text{ eV}$; :: :

The frequency of the orbital motion. (a) $112 \times 10^4 \text{ Hz}$ (b) $0.28 \times 10^4 \text{ Hz}$ (c) $5.7 \times 10^4 \text{ Hz}$ (d) $3.2 \times 10^4 \text{ Hz}$

A particle of charge $6.1 \mu\text{C}$ and mass $5.4 \times 10^{-27} \text{ kg}$ is subjected to a uniform magnetic field B directed to

velocity. Calculate the magnitude of the magnetic field required to accelerate the particle through a

200m/s (Assume the radius of the circular orbit in which it moves is 4.1cm) (a) $4.31 \times 10^7 \text{ T}$ (b) $3.5 \times$

— dad S. LOLS we Leal =

$2.7 \times 10^7 \text{ T}$ (d) $1 \times 10^7 \text{ T}$

Charged

5 hcl ta 5 'i tor carrying a

... : - of $2.7 \times 10^4 \text{ nVs}$ along the axis of a conductor

Electrically: particle is moving with a velocity of 2.7 : ine: (question

Sample Question is at a distance of 4.7mm from the center of the wire, Determine: (q

current of LIALT£ the - 7

QUITEN

a en ic : $1.07 \times 10^8 \text{ T}$ (b) $4.2 \times 10^8 \text{ T}$ (c)

Find the magnetic field intensity experienced by the particle. (a) $7.4 \times 10^7 \text{ T}$ (b) (

The magnitude of the mag

$5\% \ll A \times 9$

($31 \times 10^7 \text{ T}$) hat magnitude of force it experiences?

$4.68 \times 10^{-7} \text{ T}$ (a) same charge as an electron, What will be the mag

: rticle has the s arg:

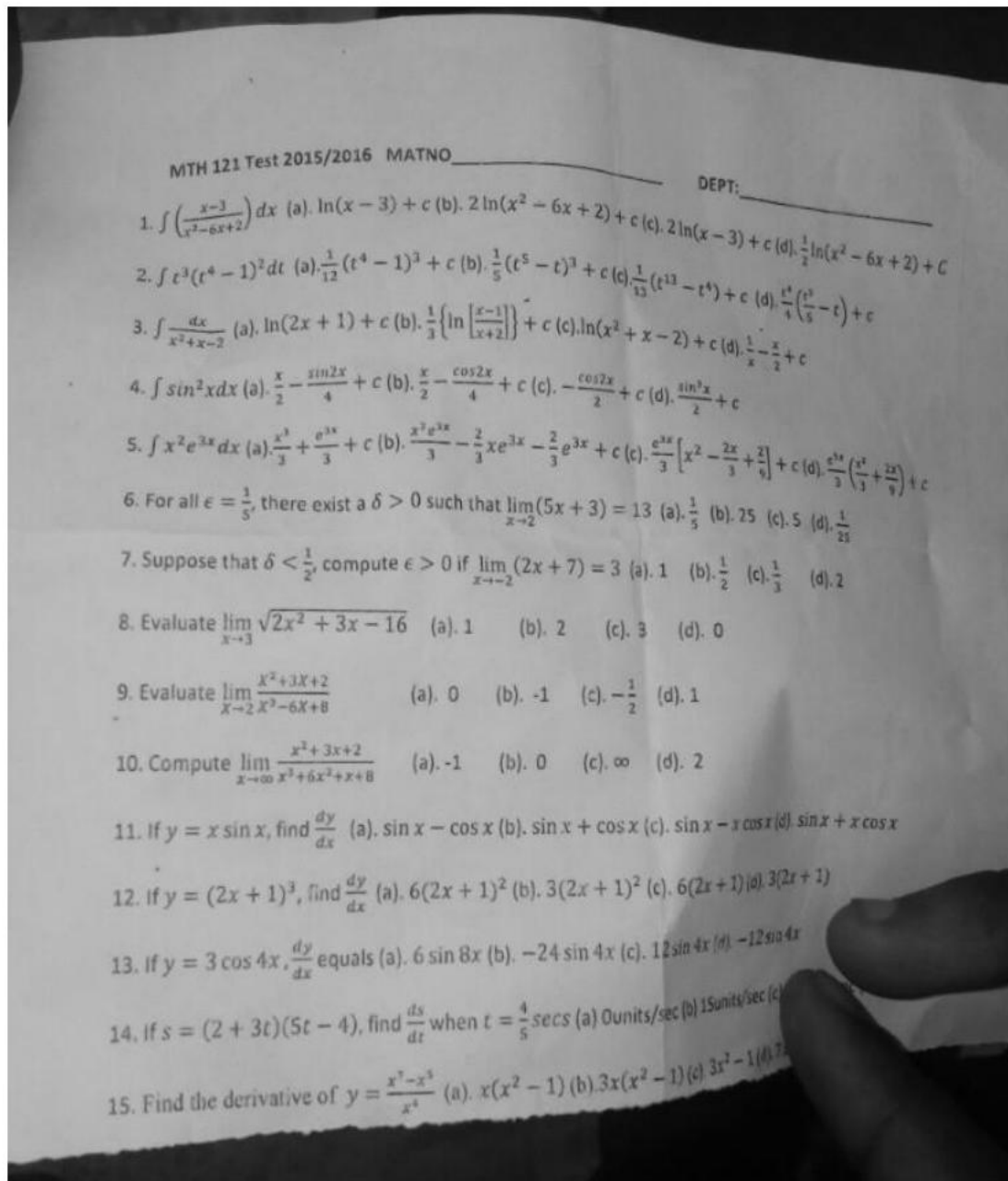
Assuming the par

aq > £02 107+" : b 1 60 072° T (c) 1 0 1077! Y (d) 20. io he) 107 N
 curren UW - R . i ' ig 4 Sem for it to produ a
 i \ : Id flow throu h a solenoid ot 330rmums and di uneter 4
 tT sho ce
 What amount of cum i
 : 762A (c) 324A (d) LISSA -
 £400mT? (a) 965A (b) 162A (©) >. sae field of $3 \times 10^7!$ *Telsa. Calculate the
 magnetic field strength of oving at a velocity of 16nv's in a MOS
 . an _ 160em ism >
 A wire of length
 _ -12yp
 a 7 68x10722V (6) 17x10
 ed in the coil. (a) TA7xLO7NV (b) 7.68x
 : induced 1n ,
 : the em.f in
 magnitude of
 ing through its
 i nount of current flowing R
 oN a has an efficiency of So. Detain (4) 715A
 7 1500w/220¥ ransformer Oma. (a) 45.5mA (b) er (6) lowing? (0) ico
 many vind Whe oad logic gate include W
 y, Hi sane Ot ina > i
 The applications of ere a =
 full adders (4) all of the aq ina full
 Sis. : (a) LA
 um is us - diode current. (8
 A diode made of an Raasev Calculate the peak
 . : ly OF 29
 sinusoidal supp)
 ders (b) multiplexers(c)
 y a resistive load of 220Q from
) 3.24 (c) 2.64 (@) 5.24

Paper 4 of 4 — PHY 102

Level 100 • First Semester • PastQuestion

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